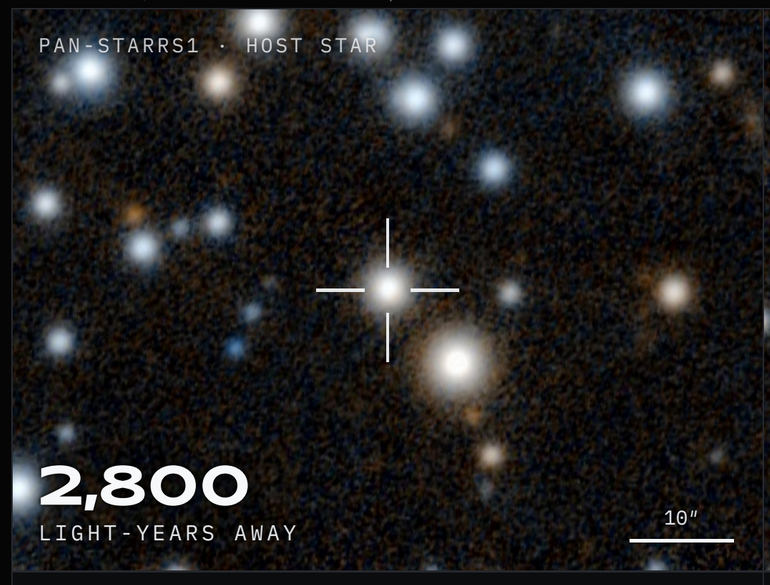
THREE SUB-NEPTUNE PLANET CANDIDATES

THREE NEW WORLDS IN THE MILKY WAY

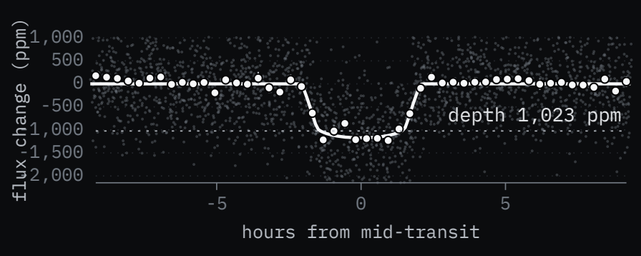
Found by Aayush Gupta in NASA Kepler data from 2009–2013, around faint stars the telescope recorded but never targeted, 2,800 to 6,400 light-years away. Each passes every test the data allow, with a false-positive probability below 1 %.[1, 2]

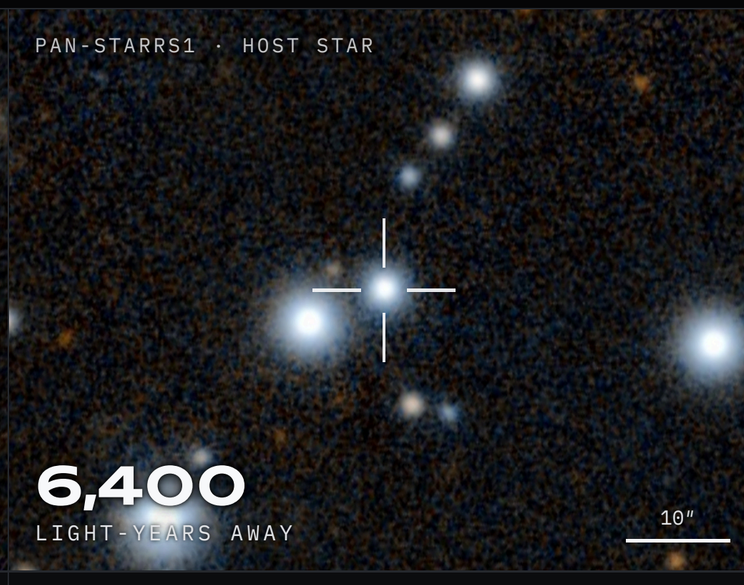
THE THREE HOST STARS, PHOTOGRAPHED, AND THE TRANSITS THAT BETRAY THEIR CANDIDATES



KIC 6699887 — 0.63 %

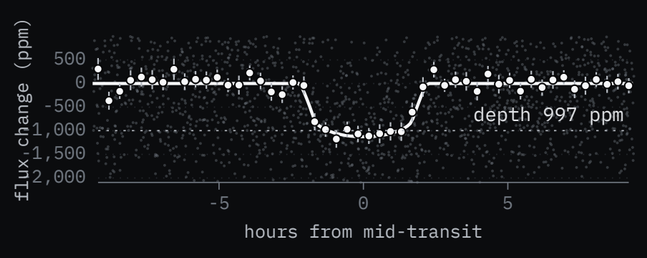
15.1136-day orbit · 2.76 Earth radii
1,023 ppm · 86 transits · T_{eq} ≈ 611 K
host 4,542 K dwarf, G 16.2, 859 pc
FPP ~~2.4 %~~ → 0.63 % with Gaia limits
second on-target signal at 47.69 d

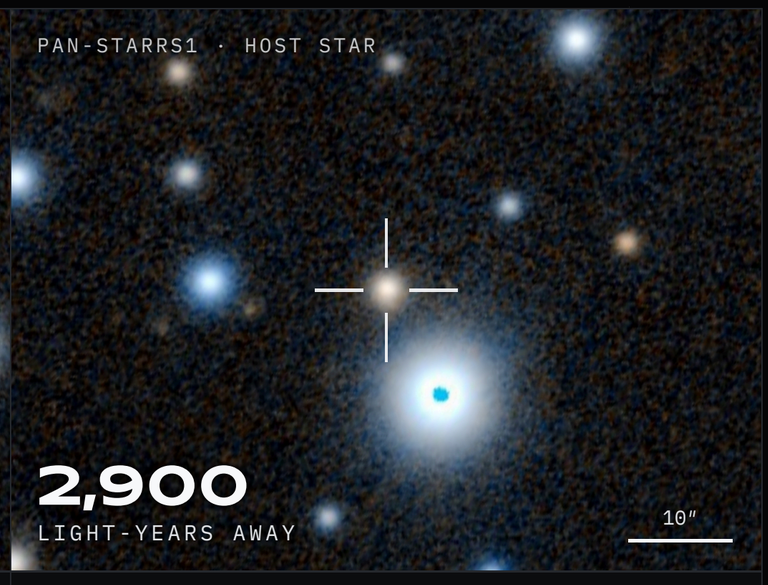




KIC 3831969 — 0.71 %

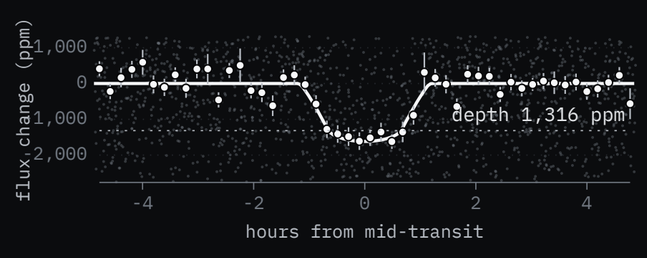
15.3133-day orbit · 3.47 Earth radii
997 ppm · 86 transits · T_{eq} ≈ 722 K
host 5,275 K dwarf, G 16.5, 1,969 pc
FPP ~~2.7 %~~ → 0.71 % with Gaia limits





KIC 11138423 — 0.70 %

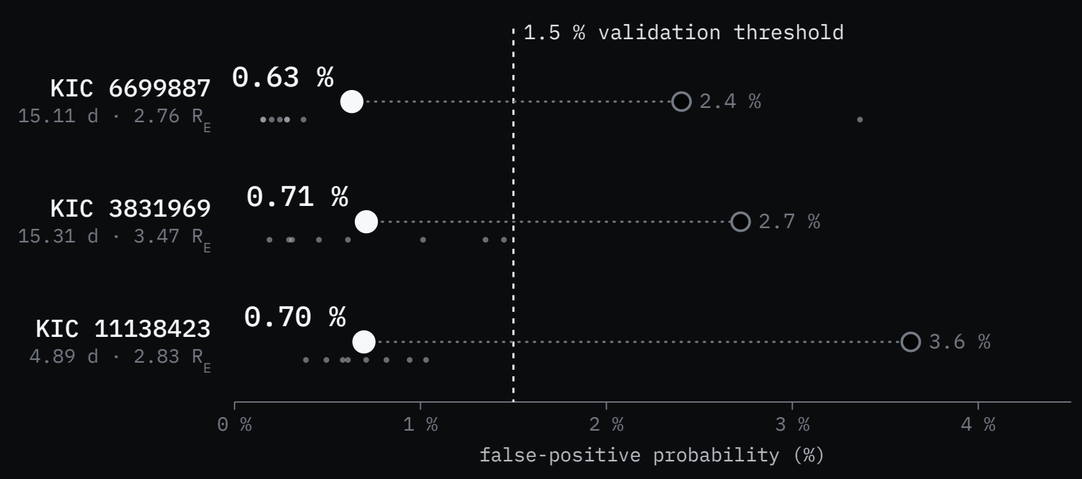
4.8945-day orbit · 2.83 Earth radii
1,316 ppm · 253 transits · T_{eq} ≈ 720 K
host 4,258 K dwarf, G 17.4, 894 pc
FPP ~~3.6 %~~ → 0.70 % with Gaia limits



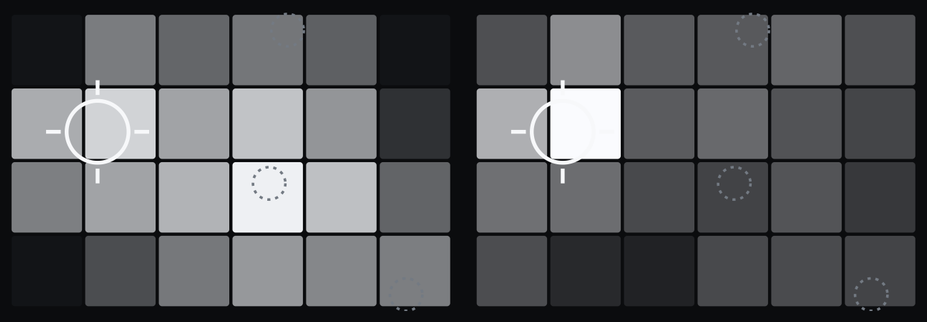
HOW THEY WERE FOUND

6,465	448	22	5	4	3
stars searched, thresholds frozen first	first-screen passes (upside-down null: 97)	pass stage-2 vetting	lose light on their own pixels	survive re-derivation and red team	candidates, FPP below 1 %

FALSE-POSITIVE PROBABILITY



THE LIGHT GOES MISSING FROM THE TARGET



KIC 6699887, Kepler season 2. Left: the stamp of the bright star Kepler observed. Right: in-transit minus out-of-transit light. The missing light sits on the candidate's Gaia position (ring): $\Delta\chi^2$ 182 vs no signal, 60 vs the next-best star.

EVERY CONDITION A CANDIDATE MUST MEET

- **Pre-registered search with a null** — Frozen target list and thresholds; the same search on every light curve turned upside down.
- **Light lost on the target's own pixels** — Two independent pixel tests, per quarter and per season.
- **Independent re-derivation** — A separate agent with no shared code recovered period and depth.
- **Transit shape fits the host star** — Stellar density from the transit matches Gaia within 0.5 σ.
- **Stage-2 vetting** — Odd and even transits match; no dip at half phase; every transit consistent; no stellar variability.
- **No known object shares the ephemeris** — Checked against every KOI, threshold-crossing event and eclipsing binary in the Kepler field.
- **Red-team review** — Agents tasked with killing each signal found no false-positive mechanism.
- **False-positive probability below 1.5 %** — TRICERATOPS with Gaia constraints: 0.63, 0.71 and 0.70 %; nearby stars 0 %.

CANDIDATE

A periodic dip that survives every test the data allow: localised on the star, absent from the inverted-flux null, unmatched by any known object, and unlikely to be anything else.

VALIDATED

False-positive probability below 1.5 % and nearby-star probability below 0.1 % in a refereed paper, usually with a high-resolution image of the host. Not yet done here.

CREDITS

Kepler Bonus photometry (Martínez-Palomera et al. 2023; doi:10.17909/7jbr-w430), Gaia DR3, TRICERATOPS, MOLUSC. Sky images: Pan-STARRS1 via CDS hips2fits, 1.2′ fields, north up; the candidates themselves are not resolved.

[1] Data & code: doi:10.5281/zenodo.23442225

[2] Research Notes of the AAS, AAS82486 (submitted)